Main.java

Share

Run

```java
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the value of N: ");

        if (!sc.hasNextInt()) {
            System.out.println("Invalid Input! Please enter an integer.");
            sc.close();
            return;
        }

        int n = sc.nextInt();

        if (n <= 0) {
            System.out.println("Invalid Input! N should be greater than 0.");
            sc.close();
            return;
        }

```

Output

Clear

```
Enter the value of N: 5
Enter 5 numbers:
1 2 3 4 5
Exception Caught: Array index is beyond the size of array.
Sum = 15

=== Code Execution Successful ===
```

Main.java

Share Run

```java
import java.util.Scanner;

class FibonacciRunnable implements Runnable {

    int n;

    FibonacciRunnable(int n) {
        this.n = n;
    }

    public void run() {

        int first = 0;
        int second = 1;

        System.out.print("Fibonacci Series: ");

        for (int i = 1; i <= n; i++) {

            if (i == 1) {
                System.out.print(first + " ");
            }
            else if (i == 2) {
                System.out.print(second + " ");
            }
            else {
```

Output

Clear

```
Enter number of terms: 5
Thread State: NEW
Fibonacci Series: 0 1 1 2 3
=== Code Execution Successful ===
```

Main.java Share Run

```java
import java.util.Scanner;

class MultiplicationThread extends Thread {

    int number;

    MultiplicationThread(int number) {
        this.number = number;
    }

    public void run() {

        System.out.println("Thread for " + number + " is Running");

        for (int i = 1; i <= 10; i++) {
            System.out.println(number + " X " + i + " = " + (number * i));
        }

        System.out.println("Thread for " + number + " is Terminated");
    }
}

public class Main {
```

Output Clear

```
Enter two numbers: 5 10
Thread 1 State: NEW
Thread 2 State: NEW
Thread for 10 is Running
Thread for 5 is Running
5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
5 X 6 = 30
5 X 7 = 35
5 X 8 = 40
5 X 9 = 45
5 X 10 = 50
10 X 1 = 10
10 X 2 = 20
10 X 3 = 30
10 X 4 = 40
10 X 5 = 50
10 X 6 = 60
10 X 7 = 70
10 X 8 = 80
Thread for 5 is Terminated
10 X 9 = 90
10 X 10 = 100
```

Main.java   Share   Run

```java
import java.util.Scanner;

class SuperClass {

    int value;

    // Parameterized constructor of Super Class
    SuperClass(int value) {
        this.value = value;
    }
}

class SubClass extends SuperClass {

    int value;   // Hides the member of Super Class

    // Constructor of SubClass
    SubClass(int superValue, int subValue) {

        // Calling Super Class parameterized constructor
        super(superValue);

        this.value = subValue;
    }

    void display() {
```

Output   Clear

```
Enter two values: 100 200
Super Class Value: 100
Sub Class Value: 200

=== Code Execution Successful ===
```

Bank.java

Share

Run

```java
import java.util.Scanner;

class Bank {

    double getRateOfInterest() {
        return 0;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Bank Name: ");
        String bankName = sc.next().toUpperCase();

        Bank bank;

        switch (bankName) {

            case "SBI":
                bank = new SBI();
                break;

            case "ICICI":
                bank = new ICICI();
```

Output

```
Enter Bank Name: SBI
SBI Rate of Interest = 8.4%

=== Code Execution Successful ===
```

RemoveDuplicates.java

Share Run

```java
            sc.close();
            return;
        }

        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            System.out.print("Enter element" + (i + 1) + ": ");
            arr[i] = sc.nextInt();
        }

        ArrayList<Integer> unique = new ArrayList<>();

        for (int i = 0; i < n; i++) {
            if (!unique.contains(arr[i])) {
                unique.add(arr[i]);
            }
        }

        System.out.println("Non-duplicate items:");
        System.out.println(unique);

        sc.close();
    }
}
```

Output Clear

```
Enter the number of elements in array: 7
Enter element1: 10
Enter element2: 0
Enter element3: 20
Enter element4: 20
Enter element5: 30
Enter element6: 30
Enter element7: 40
Non-duplicate items:
[10, 0, 20, 30, 40]

=== Code Execution Successful ===
```

SpecialCharacterCount.java

Share Run

```java
import java.util.Scanner;

public class SpecialCharacterCount {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the statement: ");
        String str = sc.nextLine();

        int count = 0;

        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            // Count special characters
            if (!Character.isLetterOrDigit(ch) && !Character.isWhitespace(ch)) {
                count++;
            }
        }

        System.out.println("Number of special Characters: " + count);

        sc.close();
```

Output Clear

```
Enter the statement: Modi Birthday @ September 17, #&$% is the wishes code for him
Number of special Characters: 6

=== Code Execution Successful ===
```

BinaryConversion.java

Share Run

```java
import java.util.Scanner;

public class BinaryConversion {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Binary Number: ");
        String binary = sc.nextLine();

        // Check if input contains only 0 and 1
        if (!binary.matches("[01]+")) {
            System.out.println("Invalid Binary Number!");
            sc.close();
            return;
        }

        int decimal = Integer.parseInt(binary, 2);
        String octal = Integer.toOctalString(decimal);

        System.out.println("Decimal Number: " + decimal);
        System.out.println("Octal: " + octal);

        sc.close();
    }
}
```

Output Clear

```
Enter Binary Number: 1101
Decimal Number: 13
Octal: 15

=== Code Execution Successful ===
```

NthLargestNumber.java   Share   Run

```java
import java.util.Arrays;
import java.util.Scanner;

public class NthLargestNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = {14, 67, 48, 23, 5, 62};

        System.out.print("Enter the value of N: ");

        // Check whether input is an integer
        if (!sc.hasNextInt()) {
            System.out.println("Invalid Input! Please enter an integer.");
            sc.close();
            return;
        }

        int n = sc.nextInt();

        if (n <= 0) {
            System.out.println("Invalid Input! N should be greater than 0.");
```

Output   Clear

```
Enter the value of N: 4
4th Largest number: 23

=== Code Execution Successful ===
```

RecursiveFactorial.java Share Run

```java
import java.util.Scanner;

public class RecursiveFactorial {

    // Recursive function to calculate factorial
    static long factorial(int n) {
        if (n == 0 || n == 1)
            return 1;
        return n * factorial(n - 1);
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the value of n: ");

        // Check for integer input
        if (!sc.hasNextInt()) {
            System.out.println("Invalid Input! Please enter an integer.");
            sc.close();
            return;
        }

```

Output Clear

```
Enter the value of n: 6
The factorial of 6 is: 720

=== Code Execution Successful ===
```

CharacterCount.java   Share   Run

```java
import java.util.Scanner;

public class CharacterCount {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int upperCount = 0;
        int lowerCount = 0;
        int digitCount = 0;

        System.out.println("Enter * to exit...");

        while (true) {

            System.out.print("Enter any character: ");
            char ch = sc.next().charAt(0);

            if (ch == '*')
                break;

            if (Character.isUpperCase(ch))
                upperCount++;
            else if (Character.isLowerCase(ch))
                lowerCount++;
            else if (Character.isDigit(ch))
```

Output   Clear

```
Enter * to exit...
Enter any character: W
Enter any character: d
Enter any character: A
Enter any character: G
Enter any character: g
Enter any character: H
Enter any character: *
Total count of lower case: 2
Total count of upper case: 4
Total count of numbers = 0

=== Code Execution Successful ===
```

AveragePositiveNegative.java

Share

Run

```java
import java.util.Scanner;

public class AveragePositiveNegative {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int num;
        int positiveCount = 0, negativeCount = 0;
        int positiveSum = 0, negativeSum = 0;

        System.out.println("Enter -1 to exit...");

        while (true) {

            System.out.print("Enter the number: ");

            if (!sc.hasNextInt()) {
                System.out.println("Invalid Input! Please
                    enter an integer.");
                sc.close();
                return;
            }

            num = sc.nextInt();
```

Output

Clear

```
Enter -1 to exit...
Enter the number: 7
Enter the number: -2
Enter the number: 9
Enter the number: -8
Enter the number: -6
Enter the number: -4
Enter the number: 10
Enter the number: 1
Enter the number: -1
The average of negative numbers is: -5.0
The average of positive numbers is: 6.75

=== Code Execution Successful ===
```

StudentGrade.java

Share Run

```java
import java.util.Scanner;

public class StudentGrade {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String[] subjects = {"Python", "C Programming", "Mathematics", "Physics"};
        int[] marks = new int[4];
        int total = 0;

        for (int i = 0; i < 4; i++) {

            System.out.print("Enter the marks in " + subjects[i] + ": ");

            // Check for integer input
            if (!sc.hasNextInt()) {
                System.out.println("Invalid Input! Marks should be integers between 0 and 100.");
                sc.close();
                return;
            }

            marks[i] = sc.nextInt();
```

Output Clear

```
Enter the marks in Python: 90
Enter the marks in C Programming: 91
Enter the marks in Mathematics: 92
Enter the marks in Physics: 93
Total = 366
Aggregate = 91.5
DISTINCTION

=== Code Execution Successful ===
```

PerfectNumbers.java

Share Run

```java
            System.out.println("Invalid Input! N should be greater than 0.");
            sc.close();
            return;
        }

        System.out.print("First " + n + " perfect numbers are: ");

        int count = 0;
        int number = 2;

        while (count < n) {
            if (isPerfect(number)) {
                System.out.print(number);
                count++;

                if (count < n)
                    System.out.print(", ");
            }
            number++;
        }

        sc.close();
    }
}
```

Output Clear

```
Enter N: 3
First 3 perfect numbers are: 6, 28, 496
=== Code Execution Successful ===
```

EmployeeBonus.java Share Run

```java
import java.util.Scanner;

public class EmployeeBonus {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the grade of the employee: "
            );
        String grade = sc.next().toUpperCase();

        System.out.print("Enter the employee salary: ");

        // Check if salary is numeric
        if (!sc.hasNextDouble()) {
            System.out.println("Invalid salary input.");
            sc.close();
            return;
        }

        double salary = sc.nextDouble();

        // Check for negative salary
        if (salary < 0) {
            System.out.println("Salary cannot be negative."
                );
```

Output Clear

```
Enter the grade of the employee: B
Enter the employee salary: 50000
Salary = 50000.0
Bonus = 5000.0
Total to be paid: 55000.0

=== Code Execution Successful ===
```

DecimalConversion.java

Share Run

```java
import java.util.Scanner;

public class DecimalConversion {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Decimal Number: ");
        String input = sc.nextLine();

        try {
            // Convert only if the input is a valid integer
            int decimal = Integer.parseInt(input);

            if (decimal < 0) {
                System.out.println("Please enter a non
                    -negative decimal number.");
            } else {
                String binary = Integer.toBinaryString
                    (decimal);
                String octal = Integer.toOctalString(decimal
                    );

                System.out.println("Binary Number = " +
```

Output

Clear

```
Enter Decimal Number: 15
Binary Number = 1111
Octal = 17

=== Code Execution Successful ===
```

PalindromeChoice.java

Share Run

```java
import java.util.Scanner;

public class PalindromeChoice {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("1. String Palindrome");
        System.out.println("2. Number Palindrome");
        System.out.print("Enter your choice: ");

        int choice = sc.nextInt();
        sc.nextLine(); // consume newline

        switch (choice) {

            case 1:
                System.out.print("Enter the String: ");
                String str = sc.nextLine();

                String revStr = new StringBuilder(str
                    ).reverse().toString();

                if (str.equals(revStr))
```

Output

Clear

```
1. String Palindrome
2. Number Palindrome
Enter your choice: 1
Enter the String: MADAM
Palindrome

=== Code Execution Successful ===
```

ATMBalance.java Share Run

```java
import java.util.HashSet;
import java.util.Scanner;

public class ATMBalance {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int totalBalance = 0;
        HashSet<Integer> used = new HashSet<>();

        int[] validDenominations = {2000, 500, 200, 100};

        for (int i = 1; i <= 4; i++) {

            System.out.print("Enter the " + i + " 
                Denomination: ");
            int denomination = sc.nextInt();

            boolean valid = false;

            // Check whether denomination is valid
            for (int d : validDenominations) {
                if (denomination == d) {
                    valid = true;
```

Output Clear

```
Enter the 1 Denomination: 500
Enter the number of notes: 4
Enter the 2 Denomination: 100
Enter the number of notes: 20
Enter the 3 Denomination: 200
Enter the number of notes: 32
Enter the 4 Denomination: 2000
Enter the number of notes: 1

Total Available Balance in ATM: 12400

=== Code Execution Successful ===
```

MthMaxNthMin.java

Share Run

```java
import java.util.Arrays;

public class MthMaxNthMin {

    public static void findResult(int[] arr, int M, int N) {

        // Check if M and N are valid
        if (M <= 0 || N <= 0) {
            System.out.println("Invalid Input! M and N should be greater than 0.");
            return;
        }

        // Sort the array
        Arrays.sort(arr);

        int size = arr.length;

        // Check if M and N are within range
        if (M > size || N > size) {
            System.out.println("Invalid Input! M or N exceeds array size.");
            return;
        }

        int mthMax = arr[size - M];
```

Output

Clear

```
1st Maximum Number = 89
3rd Minimum Number = 25
Sum = 114
Difference = 64

=== Code Execution Successful ===
```

PrimeCompositeCount.java

Share Run

```java
import java.util.Scanner;

public class PrimeCompositeCount {

    // Method to check whether a number is prime
    public static boolean isPrime(int n) {
        if (n <= 1)
            return false;

        for (int i = 2; i <= Math.sqrt(n); i++) {
            if (n % i == 0)
                return false;
        }
        return true;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int primeCount = 0;
        int compositeCount = 0;

        System.out.print("Enter the number of inputs: ");
        int n = sc.nextInt();
        sc.nextLine(); // consume newline
```

Output

Clear

```
Enter the number of inputs: 8
Enter the numbers:
4
54
63
5
89
45
32
21

Composite number: 6
Prime number: 2

=== Code Execution Successful ===
```